Deep-Sea Research II 58 (2011) 1194–1210

Contents lists available at ScienceDirect

Deep-Sea Research II

journal homepage: www.elsevier.com/locate/dsr2

Late winter oceanography off the Sabrina and BANZARE coast (117–128°E), East Antarctica

G.D. Williams [a,b,*], A.J.S. Meijers [d,b], A. Poole [d,c], P. Mathiot [e], T. Tamura [a,b], A. Klocker [b]

[a] Institute for Low Temperature Science, Hokkaido University, Sapporo, Japan
[b] Antarctic Climate and Ecosystem Cooperative Research Centre, Sandy Bay, Australia
[c] Department of the Environment, Water, Heritage and the Arts, Australian Antarctic Division, Kingston, Australia
[d] Centre for Australian Weather and Climate Research, CSIRO, Hobart, Australia
[e] Laboratoire des Ecoulements Géophysique et Industriels, Université Joseph Fourier, Grenoble, France

ARTICLE INFO

Article history:
Received 20 October 2010
Accepted 20 October 2010
Available online 5 November 2010

Keywords:
SIPEX
Under-ice
Oceanography
East Antarctica

ABSTRACT

We report on the late winter oceanography observed beneath the Antarctic sea ice offshore from the Sabrina and BANZARE coast of Wilkes Land, East Antarctica (117–128°E) in September–October 2007 during the Sea Ice Physics and Ecosystem eXperiment (SIPEX). A pilot program using specifically designed 'through-ice' conductivity-temperature-depth (CTD) and acoustic Doppler current profiling (ADCP) systems was conducted to opportunistically measure water mass properties and ocean currents at major ice stations. Additional water mass properties across the survey region were collected from Ice-Argo floats deployed during the voyage north of the 3000 m isobath. The mean drift of the floats was along the slope to the west with the Antarctic Slope Current. Vertical profiles of the potential temperature reveal the deepest ($\sim 350-400$ m) winter mixed layer (WML) in the western sector of the survey northwest of the Dalton Iceberg Tongue polynya. The meridional structure of the Antarctic Slope Front, i.e. the monotonic shoaling of the WML across the upper continental slope, is found to be similar to the previous observations in summer. A strong bottom-intensified intrusion of modified Circumpolar Deep Water (mCDW) as warm as 0 °C was detected beneath the fast ice south of the continental shelf break at 118°E. An mCDW intrusion of similar strength was detected near this location in the austral summer of 1996. We hypothesise that there is a persistent supply of mCDW and associated ocean heat flux to this region of the continental shelf that is capable of migrating to the grounding lines of the nearby Totten Glacier and Moscow University Ice Shelf. There was no detection of locally formed dense shelf water capable of forming Antarctic Bottom Water at the shelf break locations sampled despite the number of minor polynyas across this region. Ocean current measurements, limited to a maximum period of 24 h and 50–100 m depth by the relative scarcity of backscatter, found increased mean vertical speeds at the offshore stations (6–17 cm s^{-1}) relative to the shelf break (2.3–6.4 cm s^{-1}). The diurnal variation in the ADCP range reflected the diel migration of zooplankton occurring beneath the sea ice in late winter, with greater range/abundance offshore. Concurrent time series of wind, ocean current and their influence on sea ice drift from global positioning system (GPS) compass measurements were examined but the length of data acquisitions limited the applicability of this analysis.

Crown Copyright © 2010 Published by Elsevier Ltd. All rights reserved.

1. Introduction

Oceanographic observations around Antarctica are sparse due to its remote location. This is accentuated during the winter months when sea ice presents additional logistic challenges. Australian Antarctic research expeditions conducted in winter are predominantly field-based sea-ice studies, i.e. a series of 'ice-stations' ex-ship, that do not allow for standard ship-based oceanography due to time and safety constraints. For this reason a pilot field based 'through-ice' oceanography program was undertaken during the Sea Ice Physics and Ecosystem eXperiment (SIPEX) voyage (Worby et al., 2011) with the dual goals of sampling the upper water column (to 1000 m) and the surface ocean currents beneath the late winter sea ice around the East Antarctic margin. These types of measurements are not novel with similar studies previously conducted in the Weddell and Ross Seas. Leonard et al. (2006) conducted winter-long observations of ocean currents and water mass properties beneath sea ice near McMurdo Sound in 2003, focusing on the near-surface layer and the conditions for platelet ice formation.

Nonetheless, in this paper we present the first ever late winter oceanographic observations from the Sabrina and BANZARE coast of Wilkes Land between 115°E and 130°E (Fig. 1). Section 2 provides the background to this survey region, in particular the general oceanography and local regime of polynyas and ice shelves. In Section 3 we describe the two main observation platforms,

* Corresponding author at: Antarctic Climate and Ecosystem Cooperative Research Centre, Private Bag 80, Hobart, Tasmania, Australia 7001.
E-mail address: guy.darvall.williams@gmail.com (G.D. Williams).

0967-0645/$ - see front matter Crown Copyright © 2010 Published by Elsevier Ltd. All rights reserved.
doi:10.1016/j.dsr2.2010.10.035

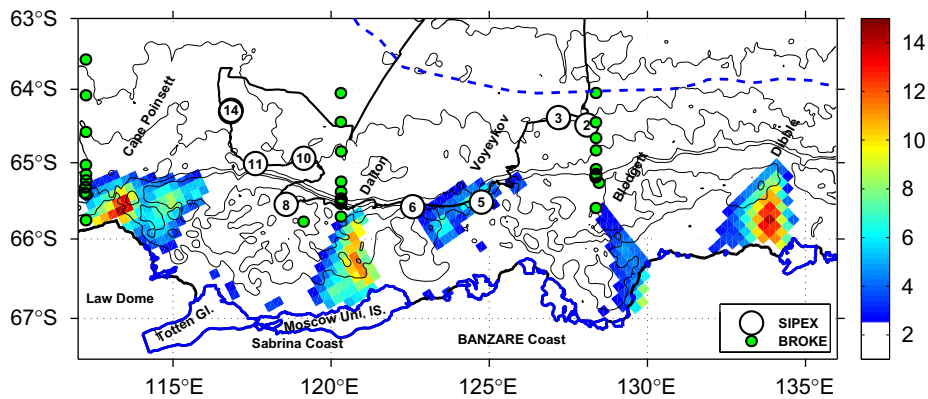

Fig. 1. The Sabrina and BANZARE coasts of Wilkes Land, East Antarctica. Bathymetric contours (−4000:1000: −1000, −500,0) from ETOPO1. Major polynya regions labelled following Massom et al. (1998) and shown with annual ice production (m/m^2) estimated from heat-flux calculation with ERA-Interim satellite data following Tamura et al. (2008). The SIPEX ship track is shown as a black line. Oceanographic stations (2, 3, 5, 6, 8, 10, 11, 13 and 14) during SIPEX shown as white circles. Stations 13 and 14 are nearly coincident. Green circles are CTD locations from the January to March BROKE survey in 1996. Transects are labelled Legs 9, 11 and 13 from west to east. The Southern Boundary from Orsi et al. (1995) is shown as a dashed blue line. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

namely a 'through-ice' conductivity-temperature-depth (CTD) and acoustic Doppler current profiling (ADCP) system, together with the ancillary data retrieved from Ice-Argo floats. In Section 4 we present our observations of the wintertime stratification and water mass distribution, with concurrent time series of ocean current, surface winds and sea-ice drift. In Section 5 we discuss the significance of the re-sampling of a warm intrusion of modified Circumpolar Deep Water with respect to local ocean/ice shelf interactions.

2. Background oceanography of the continental margin of the Sabrina and BANZARE coast

The continental shelf region of the Sabrina and BANZARE coasts of Wilkes Land has a combination of floating glaciers, ice shelves and grounded iceberg tongues, most notably the Totten Glacier and Moscow University Ice Shelf, together with a series of persistent polynya regions. These range from Cape Poinsett in the west to the Dibble Iceberg Tongue in the east (see Fig. 1) with estimates of the total sea ice production in 2007 (m m^{-2}). These sea ice production estimates come from SSM/I satellite data with the ERA-Interim reanalysis (ECMWF, 2009) and the heat flux algorithm of Tamura et al. (2007, 2008). They are all examples of recurring latent-heat polynyas which form due to ice divergence in the lee of topographic/ice barriers over the continental shelf. The exception is the Voyeykov polynya, which is a 'deep water' polynya (Morales Maqueda et al., 2004) that forms along the fast ice edge near the shelf break between 122°E and 125°E.

The previous observations in this region are limited to the BROKE (Baseline Research on Oceanography Krill and the Environment) survey in the austral summer of 1996 (Bindoff et al., 2000), as shown in Fig. 1. Vertical sections of the potential temperature for the upper 1000 m of BROKE meridional transects at 112, 120 and 128°E are shown in Fig. 2. These sections demonstrate the key water masses, boundaries and fronts in summertime and their zonal variability across the SIPEX survey region. In this paper we follow the definitions used by Orsi et al. (1995), Whitworth et al. (1998) and the subsequent update by Orsi and Wierderwohl (2009). Offshore of the shelf break, the two major water masses are cold, fresh AASW ($\gamma^n < 28.00$ kg m^{-3}) above warm, saline Circumpolar Deep Water (CDW, $28.00 < \gamma^n < 28.27$ kg m^{-3}). The CDW properties weaken polewards through mixing with cold Antarctic water masses and when cooler than 1.5 °C (also known as the Southern Boundary) it is referred to as 'modified' CDW (mCDW).

Over the upper continental slope and continental shelf break this boundary between AASW and mCDW is termed the Antarctic Slope Front (ASF, Jacobs, 1991; Whitworth et al., 1998). There is a broad westward flow, termed the Antarctic Slope Current (ASC), associated with the horizontal density gradient across the ASF, that often includes a narrow, fast-flowing jet (up to 50 cm s^{-1}) topographically pinned between the 750 and 1250 m isobaths. Though sometimes co-located, the ASC is not to be confused with the Antarctic Coastal Current (ACoC), which is associated with the East Wind Drift, is shallower and has a more coastal domain. In this paper we will concentrate on the ASC as this is typical of the circulation around the East Antarctica margin from 30°E to 150°E (Bindoff et al., 2000; Aoki, 2003; Williams et al., 2008b, 2010b; Meijers et al., 2010) and indeed much of the Antarctic margin away from the Weddell Sea and Antarctic peninsula (Fahrbach et al., 1992; Whitworth et al., 1998; Heywood et al., 1998; Thomson and Heywood, 2008).

The vertical structure of the AASW varies seasonally and is shown in Fig. 2 with a warm, fresh seasonal summer mixed layer (SML) at the surface that forms from sea ice melt. The SML deepens through the summer by wind mixing into the temperature minimum layer (T_{MIN}) which is the remnant of the previous winter's permanent mixed layer (referred to hereafter as the winter mixed layer, WML). The WML forms by convection, driven initially by atmospheric cooling of the surface SML and then by the brine rejection that occurs with sea ice formation. Inshore of the shelf break, in regions of sufficient sea ice formation/brine rejection, the WML extends to the bottom, mixing away all remnants of the summer stratification and reconnects with the base of the previous year's WML. If denser than $\gamma^n = 28.27$ kg m^{-3} and colder than $\theta = -1.7$ to -1.85 °C, this new winter water mass is referred to as Shelf Water (SW). Offshore of the continental shelf break, brine rejection is relatively weak and the WML/AASW remains fresher and lighter than the mCDW. In specific locations mCDW can upwell and migrate/intrude onto the continental shelf, transporting heat to the Antarctic coastal domain which has major implications for ocean/ice shelf interactions (Smethie and Jacobs, 2005). The mCDW intrusions are also important for the marine ecosystem if they upwell into the surface layer as they are rich in nutrients. These intrusions are also reportedly an important part of the life cycle of Antarctic krill, transporting eggs from offshore (Sala et al., 2002). However, the influence of mCDW on the continental shelf is ultimately dependent on the local stratification. For example, if SW is present, then by definition mCDW can only occupy the layer above.

The striking feature of the BROKE transects in this region was the southward migration of the warmer mCDW at 120°E, bringing

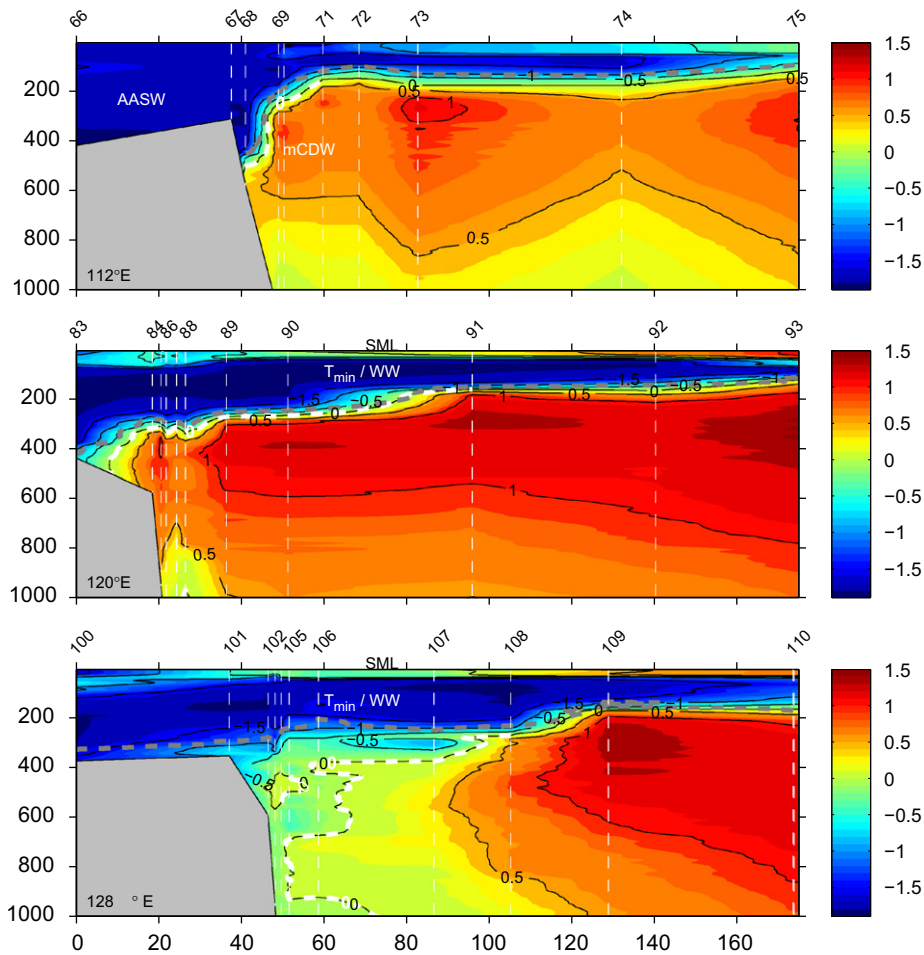

Fig. 2. Vertical sections of potential temperature (°C) in the upper 1000 m for meridional Legs 9, 11 and 13 from the BROKE survey at 112°E, 120°E and 128°E, respectively (see green circles in Fig. 1). The neutral density surface $\gamma^n = 28.00$ kg m^{-3} indicates the boundary between Antarctic Surface Water (AASW) and modified Circumpolar Deep Water (mCDW). The southern end of the 0° isotherm (dashed white line) indicates the Antarctic Slope Front, terminating once parallel with the $\gamma^n = 28.00$ kg m^{-3} contour (dashed gray line). Water masses labelled as in the text. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

the sub-surface 0 °C isotherm onto the shelf break at station 83. This was reported as the strongest mCDW signal of the entire BROKE survey between 80°E and 150°E. This is in broad agreement with the southward shift in the Southern Boundary of the Antarctic Circumpolar Current described by Orsi et al. (1995) east of 115°E (see dashed line in Fig. 1). However, the BROKE results showed that there is more zonal variability in the properties of the mCDW layer, as indicated by the southern extent of the 1 °C isotherm at 120°E in Fig. 2. The other major zonal variation across these transects was in the surface layer over the continental shelf and the weakly developed seasonal mixed layer at 112°E (i.e. it was observed north of station 72). This was not the result of any temporal lag in the sampling as the transects were conducted from east to west and instead implies greater production and/or persistence of sea ice, which has been found to delay summertime mixed layer development in other regions of East Antarctica (Williams et al., 2008b).

3. Data and methods

This project involved two independent sub-ice observation platforms: a winch-driven conductivity-temperature-depth system for measuring basic water mass properties and an acoustic Doppler current profiling (ADCP)/GPS system for measuring ocean currents and ice drift. Hereafter, these are referred to as the CTD and ADCP systems, respectively (Fig. 3). Additional water mass data were collected by Ice-Argo floats deployed during SIPEX for the University of Washington.

3.1. Ice-Argo floats from the university of washington

The Ice-Argo floats deployed during SIPEX were the new generation of Argo floats that are capable of autonomously collecting CTD and dissolved oxygen profiles beneath sea ice. These floats are designed to abort their ascent 10–30 m from the surface when sea ice cover is detected by a cold near-surface layer (Klatt et al., 2007). Deployed within the sea ice field north of the 3000-m isobath, the floats profile the water column (2–2000 m depth) every 7 days without resurfacing, using on-board data storage until the data can be uploaded after the sea ice melts. Depending on voyage logistics, the floats were either deployed from the back of the ship, through a hole in the ice during ice stations, or from a helicopter into a lead when the ship was south of the 3000-m isobath. The Ice-Argo data were extracted from the publicly available data portal operated by the National Oceanographic Data Centre (NODC, 2009). We use an abbreviated World Meteorological Organisation (WMO) naming format, i.e. WMO float number 2900130 is referred to in this paper as A-130. Data are expected to have accuracy of 0.001 °C, 0.01 and 5 m for temperature, salinity and depth, respectively (Argo, 2010). Further information about the Ice-Argo instruments and data processing can be accessed through the University of Washington (UW, 2009).

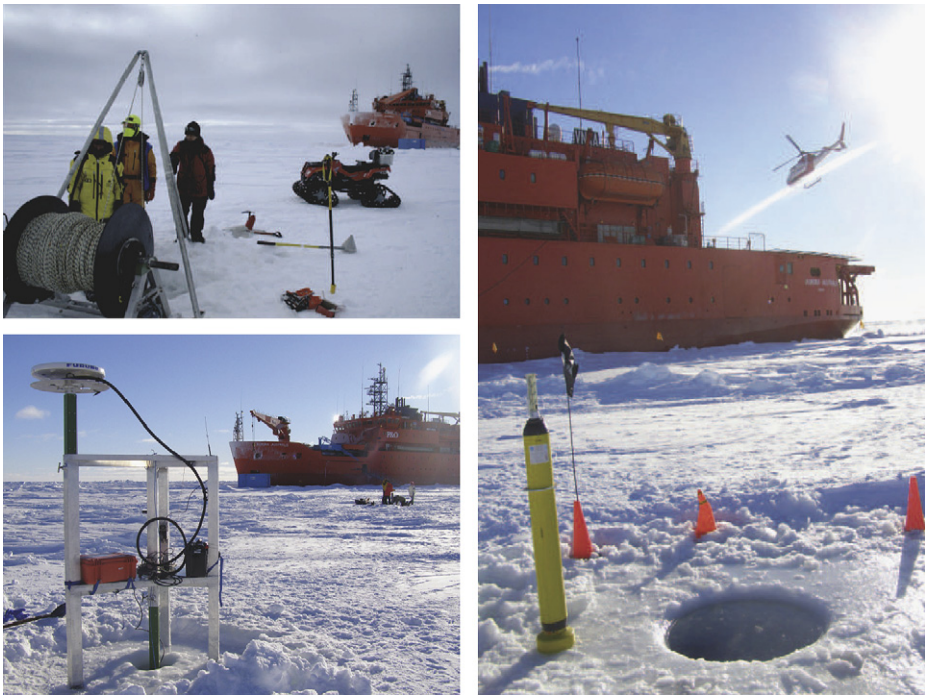

Fig. 3. Upper left panel: The SIPEX CTD system. Lower left panel: The SIPEX ADCP system. Right panel: a University of Washington Ice-Argo float and its deployment site.

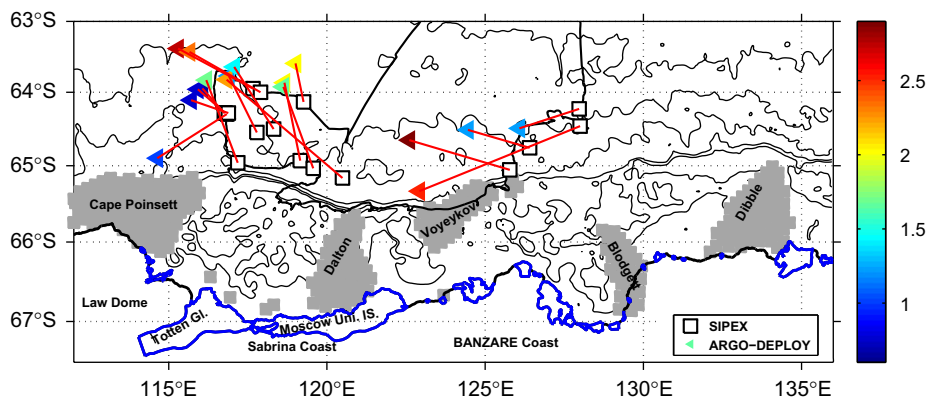

Fig. 4. Ice-Argo deployments during SIPEX. White boxes indicate the deployment location. Triangles represent the first surfacing location of each float, colored by mean drift speed (cm s^{-1}). Polynya regions from Fig. 1 shaded gray. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Fig. 4 shows the deployment locations of Ice-Argo floats during SIPEX, and their first 'download' position, with the mean speed necessary to cover this distance. The broad pattern for the time mean drift of the floats is one of along the slope to the west with the Antarctic Slope Current. The floats deployed in the western sector of the survey were transported west, but with a greater northward component, again following the bathymetry. The one exception to this was a float that surfaced near Cape Poinsett. Of course, the exact trajectories and drift speeds of these floats whilst under sea ice are unknown.

3.2. Conductivity-temperature-depth system

The CTD system comprised a Falmouth Scientific Institute (FSI) CTD instrument, a tripod and over 1000 m of polyethylene rope on a winch/drum attached to a metal sled (see Fig. 3). The winch system consisted of a control box and was powered by a portable generator. The FSI was itself housed in a protective fibreglass case to shield the sensors from rough treatment in the sub-ice environment. Initial deployments demonstrated the sensitivity of the instrument to the cold water and so the deployment procedure thereafter began by taking the instrument down to approximately 50 m and leaving it there for 3–5 min before returning to near the surface and beginning the cast. The rate of descent was kept below 1 m s^{-1} on the way down. Post-deployment the FSI was returned to the ship and the data downloaded and assessed to ensure that the data were copacetic. An interesting side-effect of the CTD system was that the 'noise' from the speed controller box effectively blocked out the ship's HF radio communications. As a result, the CTD system could not be operated when helicopter operations were being conducted within 20 km of the ship.

Overall, the conductivity/salinity data collected by the FSI were poor (for further details, see Rosenberg, 2009). In the later stages of SIPEX, a single shipboard CTD (Seabird SBE911+ CTD) profile was conducted with the FSI attached for calibration. The vertical

profiles from this cast are shown in the top row of Fig. 5 and clearly show a large offset in salinity (> 0.1). The first profile from the nearest Ice-Argo float (A-120) indicates that the float performed better than the FSI, closely matching the Seabird profile. Simultaneous data collected during bottle stops on the upcast was used to determine linear best fits for pressure, conductivity and pressure. An attempt to calibrate all FSI profiles using the combined FSI/Seabird result was conducted with the results again compared to nearby Ice-Argo profiles. This appeared successful for Station 2 (float A-130), in conjunction with a nearby profile from BROKE (Stn 109), but not thereafter, with Stations 10 and 11 showing the 'calibrated' salinity data as still much fresher than at the nearest Ice-Argo data (floats A-114 and A-125, respectively). Similar residual offsets in salinity post-calibration were found when the calibrated salinity profiles at Station 8 were compared with nearby historical salinity data from BROKE (Stn 82). These offsets in salinity are simply too large, in particular at depths below the seasonally variable surface layer, for this to be accounted for by spatial or temporal variability.

Upon completion of the SIPEX voyage, further testing of the FSI instrument revealed that the conductivity sensor was highly sensitive to the protective fibreglass casing it was deployed in, and that minor movements (1–2 mm) in the orientation of the instrument relative to the casing during battery changes were likely responsible for the offset and variable behaviour during the voyage. For this reason we dismiss the absolute value of salinity from this instrument, but present it here nonetheless for its vertical structure, in particular for the stations on the shelf break (lower panel, Fig. 5) that were not sampled by the Ice-Argo floats.

3.3. Acoustic Doppler current profiling system

The acoustic Doppler current profiling (ADCP) system was designed and constructed by Marine Science Support at the Australian Antarctic Division with the goal of observing the ocean current regime beneath the sea ice. To determine the absolute ocean current, a GPS compass was included in the system to account for sea ice drift. The system consisted of a three-legged supporting platform that included a central, downward facing retractable pylon for the RDI Workhorse ADCP operating at 307.2 kHz, a flat tray for storing the power supply/control system, mounting points for the radio modem antenna and an upward facing extendable pylon for the Furuno SC50 GPS compass (see Fig. 3). The observation site was preferentially in the zone perpendicular to the port bow of the ship, between 50 and 100 m away to minimise any contamination of the flow field from the ship's bow thrusters. The standard ship-mounted ADCP of the R.V. *Aurora Australis* was not functional during the SIPEX voyage.

At each station a hole through the sea ice was prepared and the ADCP lowered up to 1.5 m, being fully submerged and clear of the base of the ice. The ADCP, GPS and radio modem were connected to the 24 V rechargeable portable battery through a control box. The system was designed so that the Furuno GPS compass and ADCP remained in a fixed position relative to each other, and the entire system was positioned parallel to the ship's heading. The basic procedure began with the initialisation of the GPS compass and establishing of communication with the ADCP computer on board the ship via the radio modem. Thereafter, the ADCP was initialised from the ship using the RDI WinRiver software. After initial testing determined a lack of available backscatter in the water column, a common problem in Antarctic waters accentuated by the scarcity of biological matter in late winter, the sampling bin size was set to 8 m to achieve a maximum range of 50–100 m with a 1.76 m blanking window below the transducer head. This subsequently ruled out any small-scale sampling of the immediate boundary layer below the ice. The ADCP collected single ping ensembles every 0.5 s, operating using the RDI default settings.

The ADCP data were post-processed during the voyage using the RDI WinRiver software. This synchronised the GPS heading data with the ADCP velocity data streams, producing velocity headings relative to true north and the ADCP position, and removed bad data points where data were below the beam correlation threshold or the water velocity exceeded the threshold of 107 cm s^{-1}. The ice-mounted ADCP frame ensured that ADCP tilt was fixed near 0° and all four ADCP beams were required to produce solutions. Ensemble averaging over 3 min (360 pings) reduced the RMS error to below 0.5 cm s^{-1} for all stations except station 3, where it is below 1 cm s^{-1}. The water velocities relative to the ADCP were converted to absolute u and v velocities by adding the ADCP (ice) drift velocity components calculated based on the 90 s boxcar filter smoothed GPS compass position. All stations exhibited strong vertical

Table 1
ADCP station current statistics. Eastward and northward velocity components (u and v), the absolute speed (S), maximum absolute speed (Max), the standard deviation (σ), the temporal mean (overbar) and the vertical mean (< >). σ_{head} is the standard deviation of the compass heading at each bin from the water column mean heading at that time. Subscript t indicates tidal velocities from the Circum-Antarctic Tidal Simulation, vers. 2008b. Ice thickness (Z_{ICE}) from Worby et al. (2011). Velocities are in m s^{-1}, depths and ice thickness are in m, headings are in degrees and durations are in decimal days.

Station	Depth	Duration	$\overline{<\mathbf{u}>}$	$\sigma<\mathbf{u}>$	$\overline{<\mathbf{v}>}$	$\sigma<\mathbf{v}>$	$\overline{<\mathbf{S}>}$	$\sigma<\mathbf{S}>$
1	3050	0.17	−0.04	0.03	0.1	0.04	0.11	0.04
2	2799	0.96	−0.07	0.06	0.04	0.07	0.11	0.04
3	3214	0.15	−0.33	0.05	0.11	0.05	0.35	0.05
5	448	0.76	−0.01	0.01	0.02	0.02	0.03	0.01
6	755	0.19	0.06	0.03	−0.02	0.01	0.06	0.03
8	588	0.48	0.02	0.01	0.01	0.02	0.03	0.01
10	2898	1.01	−0.02	0.03	−0.04	0.07	0.07	0.06
13	2054	0.25	−0.11	0.01	0.08	0.03	0.14	0.01
14	2566	1.01	−0.01	0.10	0.01	0.18	0.18	0.1
Station		$<Z_{ICE}>$	σZ_{ICE}	**Max($\bar{S}$)**	$\sigma_{\bar{S}}$	σ_{head}	$\overline{<\mathbf{S_t}>}$	**Max(S_t)**
1		0.81	0.54	0.21	0.01	11.91	0.01	0.02
2		1.03	0.37	0.31	0.02	6.62	0.01	0.02
3		0.99	0.44	0.49	0.05	5.27	0.01	0.01
5		0.71	0.13	0.18	0.01	40.24	0.03	0.06
6		1.15	0.67	0.11	0.02	21.88	0.01	0.01
8		0.59	0.43	0.07	0.01	37.1	0.02	0.04
10		0.70	0.46	0.27	0.01	12.26	0.02	0.04
13		1.04	0.45	0.19	0.01	5.71	0.01	0.02
14		1.14	0.76	0.53	0.01	6.79	0.01	0.02

coherence in velocity with standard deviations of shear of less than 1–5 cm s^{-1} absolute velocity (Table 1). Because of the relatively narrow range of depths sampled and the strong vertical coherence observed, we use the calculated vertical mean when referring to horizontal components of velocity.

3.4. Ocean tides

As the ADCP time series were too short (maximum of 24 h) to meaningfully estimate the tidal contribution to currents through frequency analysis, tidal velocities were estimated using the Circum-Antarctic Tidal Simulation (CATS - vers. 2008b, Padman et al., 2002). Subtracting the tidal predictions from the ADCP data only successfully reduced the variance of the u and v components of velocity at stations 2, 8, 13 and 14, suggesting that the phase of the tide model is inaccurate in the SIPEX region. This inaccuracy may also be caused by non-tidal currents occurring in phase with the tidal signal, but without a longer time series of observations this cannot be resolved. As we cannot unambiguously separate out the tidal components we only use the tide model as a gauge of the tidal velocity magnitudes in this analysis and leave the observations unmodified. All velocities discussed in this paper therefore include a tidal component. Table 1 shows that north of

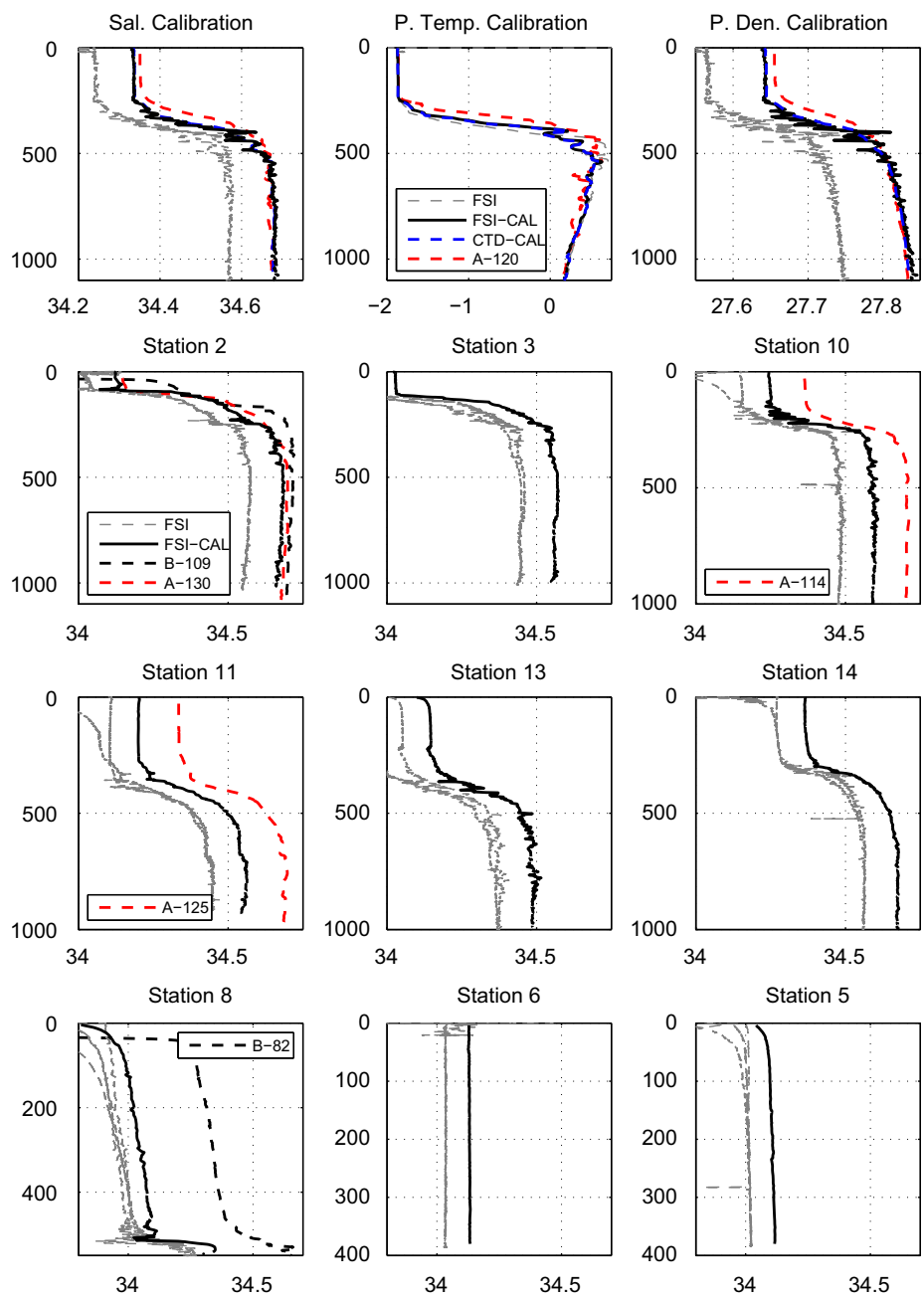

Fig. 5. SIPEX salinity data. Top row: calibration results (black) between FSI (gray—downcast and upcast, some multiple profiles) and shipboard Seabird SBE-11 (dashed blue). Also shown is a salinity profile from Ice-Argo (float A-120). Rows 2–4 show the raw (gray) and calibrated FSI salinity (black) at SIPEX oceanographic stations (see Fig. 1). At stations 2 and 8, the nearest available profile from BROKE (stations 109 and 82, respectively) is shown as a dashed black line. At stations 2, 10 and 11, the nearest available profile from Ice-Argo (floats A-130, A-114 and A-125, respectively) is shown as a red dashed line. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

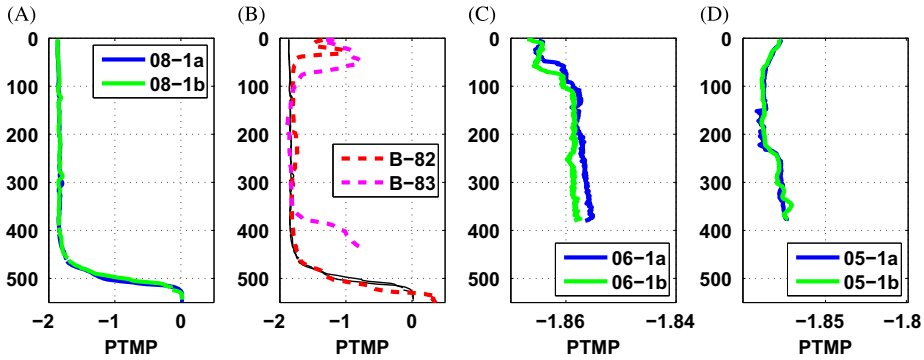

Fig. 6. Vertical profiles of potential temperature (°C) at stations 8, 6 and 5 (panels A, C, D, respectively) and BROKE stations 82 and 83 (panel B) in the vicinity of SIPEX station 8 on the continental shelf break (see Fig. 1). In each case the upcast from double deployments (blue and green lines) are shown. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

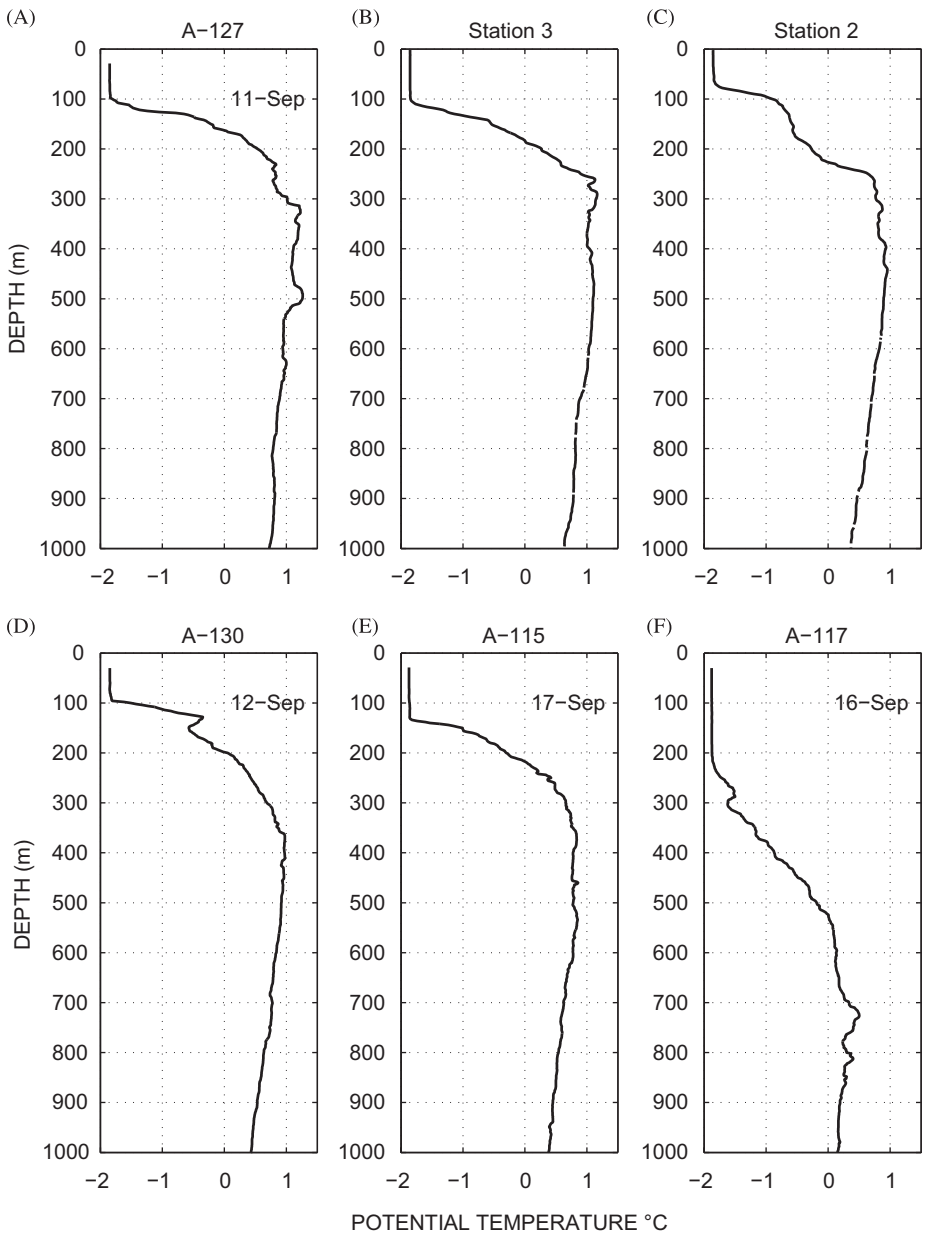

Fig. 7. Vertical profiles of potential temperature (°C) from SIPEX ice stations (2 and 3) and the deployment profiles of Ice Argo (float numbers 127, 130, 115 and 117) in the eastern sector of the survey. Panels A–F depict locations from north to south.

the shelf break tidal velocities are approximately an order of magnitude smaller than the ADCP mean current absolute velocities, while over the shelf at the shallower stations, the tidal currents are of a similar magnitude to the mean ADCP velocity and may significantly influence it.

4. Results

4.1. Water mass properties from 'through-ice' CTD and Ice-Argo

We begin with the water mass properties observed during SIPEX and focus on two main results: the detection of modified Circumpolar Deep Water on the continental shelf across fast-ice stations 5, 6 and 8, and the large-scale vertical stratification, i.e. the winter mixed layer depth, across the offshore stations and Ice-Argo profiles.

4.1.1. Modified Circumpolar Deep Water on the continental shelf north of the Totten Glacier

After sampling two vertically homogeneous profiles of near-freezing temperature water at shelf break stations 5 and 6, warm mCDW was detected at station 8. Figs. 5 and 6 show that the bottom layer of station 8 had a 50–70 m layer of mCDW that is warmer (up to 2 °C) and more saline (at least 0.2) relative to the water column above. This is consistent with the mCDW layer previously recorded in summertime during BROKE at station 82. The strength of the warming signal implies that the Antarctic Slope Front is on or south of the shelf break, as in the western Ross Sea (Whitworth and Orsi, 2006; Muench et al., 2009). That this feature was observed during SIPEX in late winter, more than 10 years after the summertime observation during BROKE, implies that this intrusion is likely to be a persistent feature.

In other regions of East Antarctica, most notably the Adélie and George V Land continental shelf break, mCDW intrusions are much weaker in terms of their warming signal relative to the 0 °C

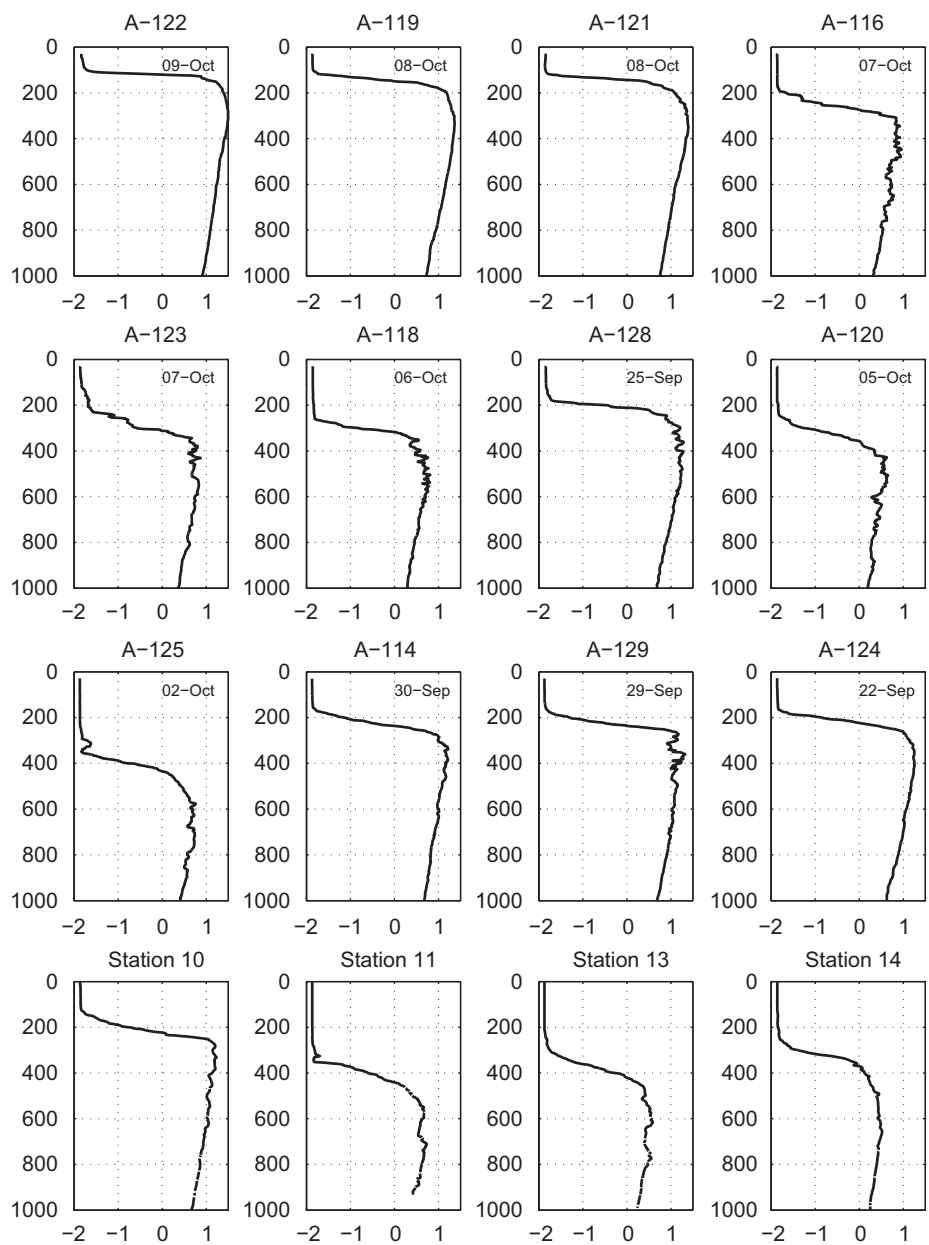

Fig. 8. As in Fig. 7, but for the western sector of the survey at SIPEX stations 10–11, 13–14 (bottom row) and the deployment profiles for Ice Argo floats A-114, A-116, A-118–125, A-128–129.

observed here (Williams et al., 2010a), in particular during winter when the mCDW is almost unidentifiable due to strong convective mixing during dense shelf water formation (Williams and Bindoff, 2003). Additionally, the mCDW intrusions for the AGV region are lighter than the local shelf water and penetrate the continental shelf water column at mid-depths (300–400 m). This is important because this means the heat flux associated with the mCDW cannot directly influence ocean/ice shelf interactions at the grounding lines (> 1000 m, Rignot and Jacobs, 2002) of the Mertz and Ninnis Glacier in that region. We note that for the Sabrina coast the mCDW intrusions observed during both BROKE and SIPEX occupy the bottom layer of the observed water mass profiles. This is significant for two reasons. It suggests firstly that there was no dense shelf water present in this location at the time of sampling and secondly that when mCDW is the densest water mass, it can influence a greater depth range of the nearby Totten Glacier which has an even deeper grounding line than the Mertz and Ninnis glaciers (Rignot and Jacobs, 2002).

4.1.2. Winter mixed layer depth over the continental slope and rise

As introduced earlier, the winter mixed layer depth (WML) defines the base of the AASW layer and is the product of the current season's sea ice production and the local mesoscale circulation on the continental shelf. It is observed to exhibit a meridional structure from summertime surveys, shoaling north of the upper continental slope to form the horizontal density gradient of the Antarctic Slope Front and generate the baroclinic Antarctic Slope Current. One of the goals of SIPEX was to firstly determine if this meridional structure exists immediately after the sea ice production season, and secondly to examine whether any large-scale variability in the winter mixed layer depth could be attributed to the spatial variability of sea-ice production across the survey region.

We separate the sampling stations into eastern and western sectors of the SIPEX survey and start by presenting vertical profiles of the potential temperature for the eastern sector in Fig. 7. The WML is relatively shallow (~ 100 m) at the northern stations, i.e. A-127, FSI-CTD stations 3 and 2 and then there is a transition to the deeper WML (from ~ 100 to ~ 200 m) across the Antarctic Slope Front between the initial profiles for Ice-Argo float A-130 and A-115 to A-117, northeast of the Voyeykov polynya. The profile from station 2 is unique in that there is a double-step across the permanent thermocline separating the base of the WML and the mCDW below.

The vertical profiles from the western sector of the survey (see Fig. 8) show a deepening of the WML depth, relative to the eastern sector, to more than 300 m. The deepest WML are at FSI-CTD stations 11, 13 and 14. The spatial variability in both WML depth and sub-surface T_{max} below the WML during late winter and summer from SIPEX CTD and Ice-Argo, respectively, is shown in Fig. 9. There is a shift to shallower WML depths from south to north, similar to that observed in summer (Fig. 2), in relation to the meridional barotropic fields associated with the Antarctic Slope Current. The region of deepest WML, extending furthest offshore from the shelf break is found between 115–120°E. This is northwest of the Dalton Iceberg Tongue polynya region and suggests a link to the ice production therein. However the northern extent of the 'deep' WML is more likely due to the northward shift in the

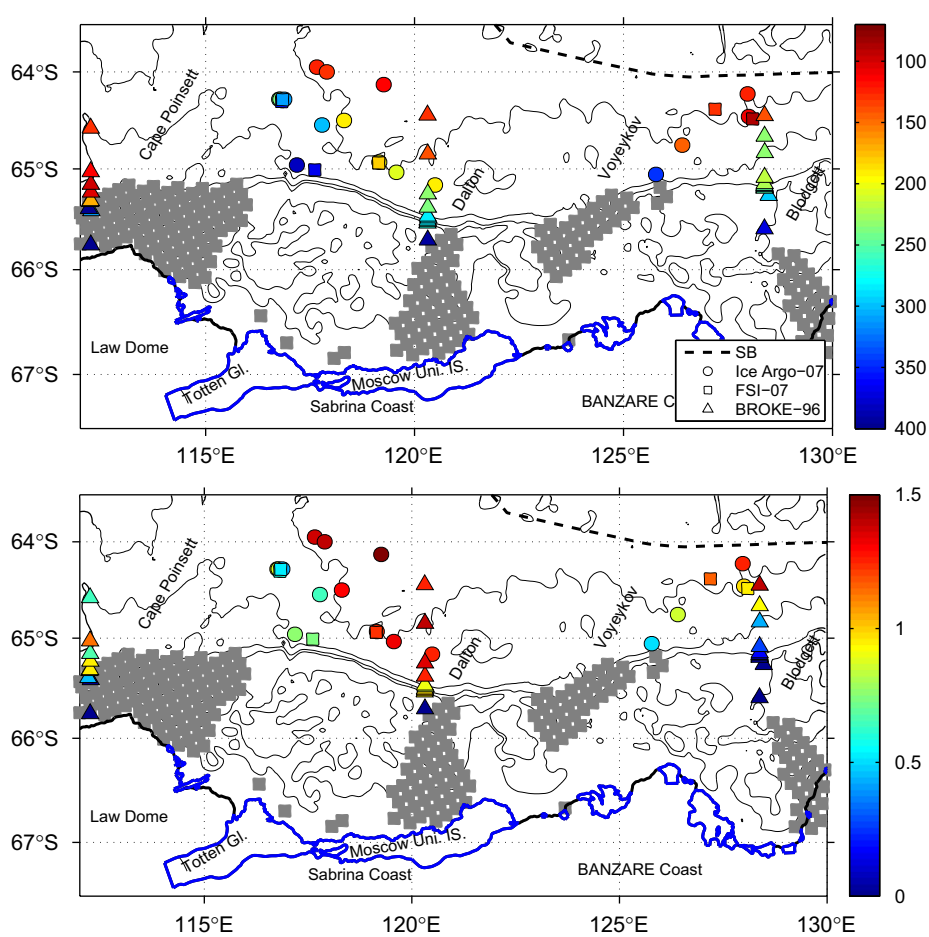

Fig. 9. Upper panel: winter mixed layer depth from SIPEX FSI (squares), Ice Argo (circles) and BROKE (triangles). Lower panel: temperature maximum of mCDW from SIPEX FSI (squares), Ice Argo (circles) and BROKE (triangles). Polynya regions shaded gray.

bathymetry in that region deflecting the Antarctic Slope Current to the north-west. The spatial distribution of sub-surface T_{max} below the WML also clearly indicates the southern migration of the warmest mCDW/CDW east of this region at 120°E. Ultimately, the lack of true repeat sampling between BROKE and SIPEX disallows definitive comparison of seasonal and decadal variation in the water mass properties and boundaries.

4.2. Ocean currents from the 'through-ice' ADCP system

4.2.1. Offshore stations

Of the six major ice stations executed north of the shelf break, over bathymetry of 1500–4000 m, three (2, 10 and 14) were conducted for periods close to 24 h, while the remaining three (1, 3 and 13) all lasted for less than 7 h. All stations are highly coherent in the vertical with RMS differences between the mean heading and the heading at individual bin depths being between 5° and 12° for all stations. The influence of tides is small north of the shelf break (mean magnitude of order 1 cm s^{-1}, Table 1) and the current behaviour is not significantly modified by the removal of model estimated tidal components.

The column mean speed and heading of the water beneath the sea ice at the long period stations (2, 10 and 14) are shown in Fig. 10. The top 100 m below the sea ice tends to be relatively fast (mean magnitude 6–17 cm s^{-1}), with maximum speeds of over 30 cm s^{-1} at stations 2 and 14. At these long period stations the currents vary strongly on time scales of a few hours in both magnitude and direction. This is particularly apparent at stations 10 and 14 where the flow reverses direction over a period of less than five hours. Similar sharp changes in flow direction occur at station 3, although the change in heading is not as dramatic. In almost all cases these changes occur over relatively short times and are separated by longer periods where the heading and velocity are relatively steady. This suggests that tidal or inertial oscillations (both of which have periods of the order of 12 h at this latitude) are not responsible for these direction changes. We examine the role of the wind in driving the observed current heading shifts in the next section.

A mean westward velocity component (ranging between 3 and 33 cm s^{-1} with variances of 1.3–9.7 cm s^{-1}, including tides, Table 1) is observed at all of the offshelf stations, which is consistent with other hydrographic observations (Bindoff et al., 2000) and the presence of the ASC north of the shelf break. Periods of eastward velocities do occur at stations 3, 10 and 14, but with the exception of station 14, all are relatively brief. Less consistency is apparent in the meridional component (variance 2.6–18.1 cm s^{-1}) and the long period stations all exhibit periods of both northward and southward velocities with magnitudes greater than 10 cm s^{-1}. However, the northward current tends to dominate and the resultant net flux over the observed periods is northwestward at all stations except station 10, where it is southwestward (mean meridional velocity -4 cm s^{-1}).

4.2.2. Fast ice stations near the continental shelf break

The stations occupied over the shelf (<800 m) were distinctly different from those north of the shelf break (see Fig. 11). Smaller currents were encountered (2.3–6.4 cm s^{-1}) and tidal velocities estimated using the CATS2008b tide model were stronger, with mean magnitudes of up to 3 cm s^{-1} and peak velocities of over 6 cm s^{-1}. The current direction was also significantly more variable and the variability was more continuous than offshore, where there had been periods of coherence punctuated by shorter bursts in variation. While offshelf currents were broadly northwestward, there was no consistent current heading between the three onshelf

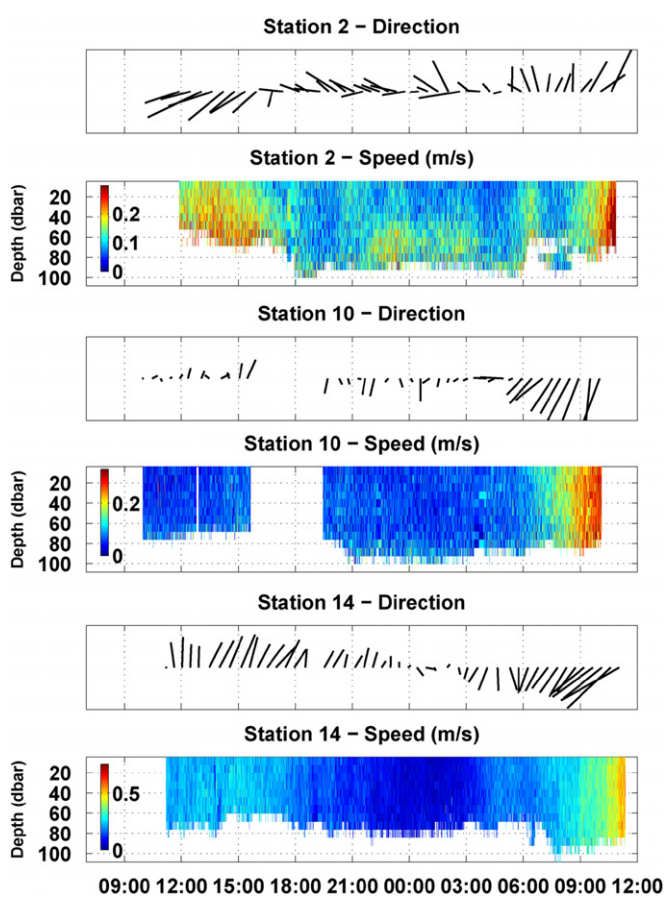

Fig. 10. ADCP direction and speed (m s^{-1}) at offshore SIPEX ice stations 2, 10 and 14.

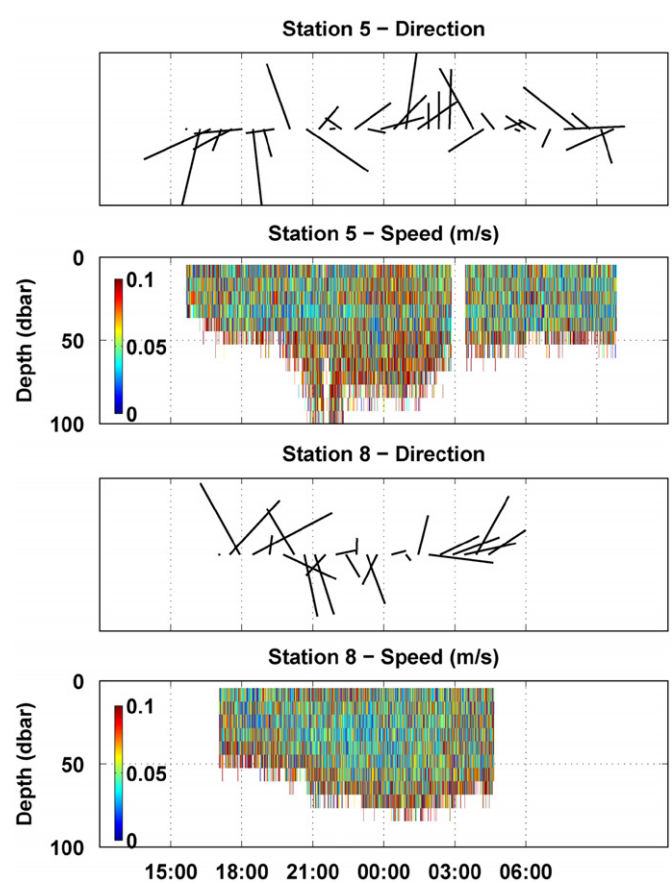

Fig. 11. As in Fig. 10, but for SIPEX stations 5 and 8 on the continental shelf.

stations. Additionally, the onshelf current meters are significantly less vertically coherent than those offshelf, and the standard deviation of the vertical water column heading was 22–40°. This apparent reduction of vertical coherence is a result of the generally weaker mean currents and hence reduced signal to noise ratios over the continental shelf. The mean μ/σ ratio where μ is the vertical mean absolute velocity and σ the standard deviation is <0.15 north of the shelf break and increases to >0.35 over the shelf. The high variability and potential for aliasing introduced by the tides and short observational period means that the mean field cannot be meaningfully inferred from these data. However it does illustrate the significant change in flow regime across the continental shelf break and the relative importance of tides south of this.

4.2.3. ADCP range as proxy for the diel migration and relative abundance of zooplankton during SIPEX

A biological sub-program of the SIPEX project sampled Antarctic Krill (*Euphasia superba*) populations beneath the sea ice (O'Brien et al., 2011). Previous studies have used ADCP backscatter to detect zooplankton beneath sea ice (Cresswell et al., 2009; Kaneda et al., 2002). For the SIPEX results, the clear diurnal variation in ADCP range invites a qualitative assessment, albeit extremely simplified, of the diel migration of zooplankton beneath sea ice in this region in late winter. Fig. 12 shows the diel variation in ADCP backscatter and relative distribution of maximum ADCP range across the SIPEX survey. If we assume that ADCP range is directly proportional to the presence of zooplankton, in particular in late winter before the sea ice melts and contributes detritus to the backscatter of the water columns, then we find that diel migration of zooplankton was occurring beneath sea ice in late winter. That is, the krill migrated away from the surface during the day, reducing the ADCP range and then returned to the surface layer in the evening, increasing the ADCP range to its maximum near midnight. The presence of zooplankton was confirmed by direct observations/sampling through the ice holes and camera surveys conducted after each 'through-ice' CTD (pers. comm. P. Virtue). If we make the further assumption that the species of zooplankton and associated acoustic backscatter was constant, then we also find that there was greater abundance of zooplankton at the offshore stations relative to the stations on the shelf break beneath the fast-ice.

4.3. Wind, ocean current and sea ice drift

Sea ice drift is the result of the summation of the forces acting upon it, most notably the wind, ocean currents, internal ice dynamics (depending on local sea ice concentration) and sea surface tilt (Leppäranta, 2005). On daily to weekly timescales sea ice and the surface layer of the ocean will predominantly be wind forced, however, in certain circumstances the ocean forcing can dominate the wind. The GPS compass used in the ADCP system to determine the absolute ocean current provides highly accurate measurements

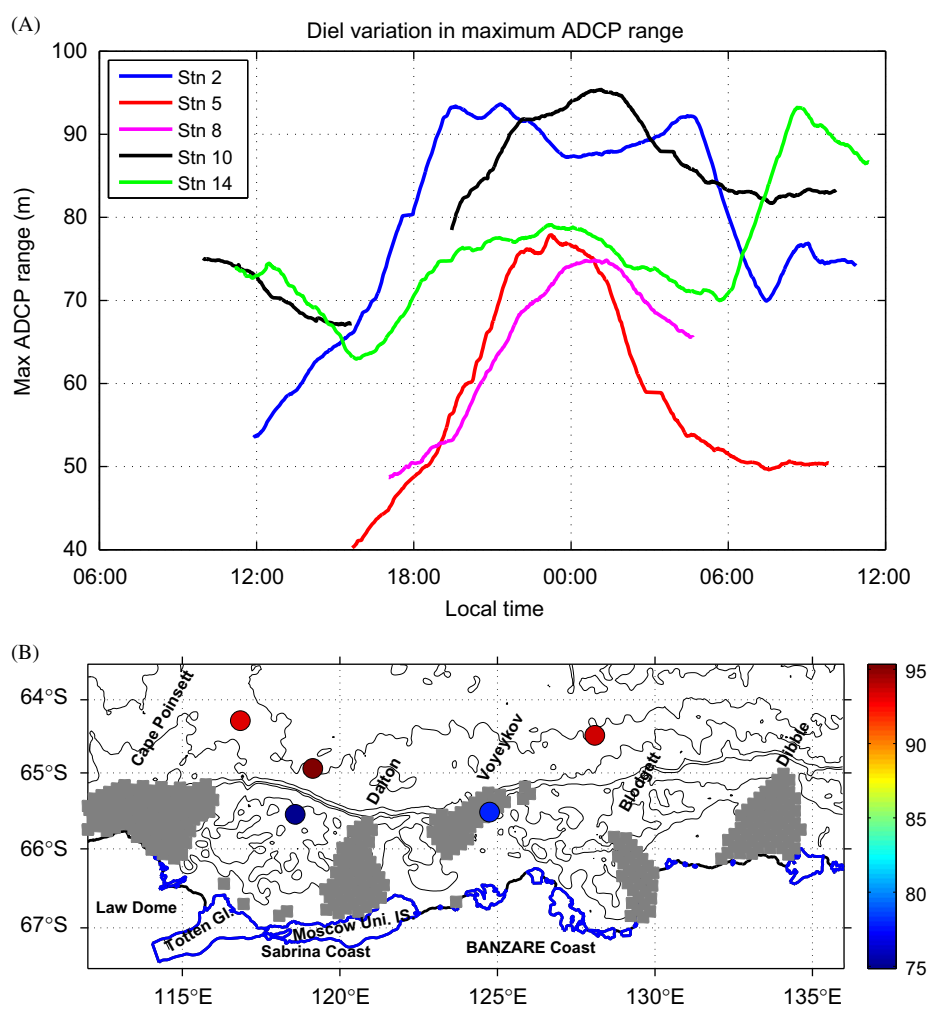

Fig. 12. Panel A. Time series of xmaximum range (m) of the ADCP as proxy for diel migration of zooplankton. Panel B. Spatial variability in the maximum ADCP range across the SIPEX survey.

of sea ice drift. Simultaneous measurements of wind from the underway system on board the R.V. *Aurora Australis* present us with an opportunity to make a preliminary assessment in this paper on the relative influence of wind and ocean currents on the sea ice drift measured during the SIPEX ice stations. The wind data presented here were taken as 10 min averages collected outside of the ship's wind shadow at $\sim$34 m height and recalculated for the $\sim$15 m height using Prandtl's logarithmic law (von Karman's constant K=0.4 and roughness length Z_0=0.02) following Blackadar (1982). The heading convention used here is the standard oceanographic one, i.e. the wind is moving toward the indicated heading.

In Fig. 13 we present the speed and 'adjusted' heading for ocean current, wind and ice drift at the longer period stations (2, 10 and 14). The headings shown here are plotted to ensure that there are no discontinuities across the 359 $\Rightarrow$ 0° transition. There are clearly strong correlations between the components of the ice/ocean/wind system at each station, with the speeds of all three varying together closely. Additionally, there is also an obvious dependence between the heading of the ice and the headings of the wind and ocean. However, it is not always clear whether the wind or ocean is dominant in driving the ice heading, or that the wind dictates the surface ocean heading. For example, the ice and ocean headings are almost exactly matched at station 2, while the wind heading is almost 100° counterclockwise of these, in the opposite sense to the expected Ekman rotation. In a counterexample, the ice and wind heading are closely matched at the beginning of the station 10 time series, but suddenly diverge after 11.00 pm and the ocean heading appears to drive the ice after this. The ocean and wind rotate in opposite directions at station 10 from 8.00 pm to 12:00 am, as well as at station 14 between 11:00 pm and 2.00 am.

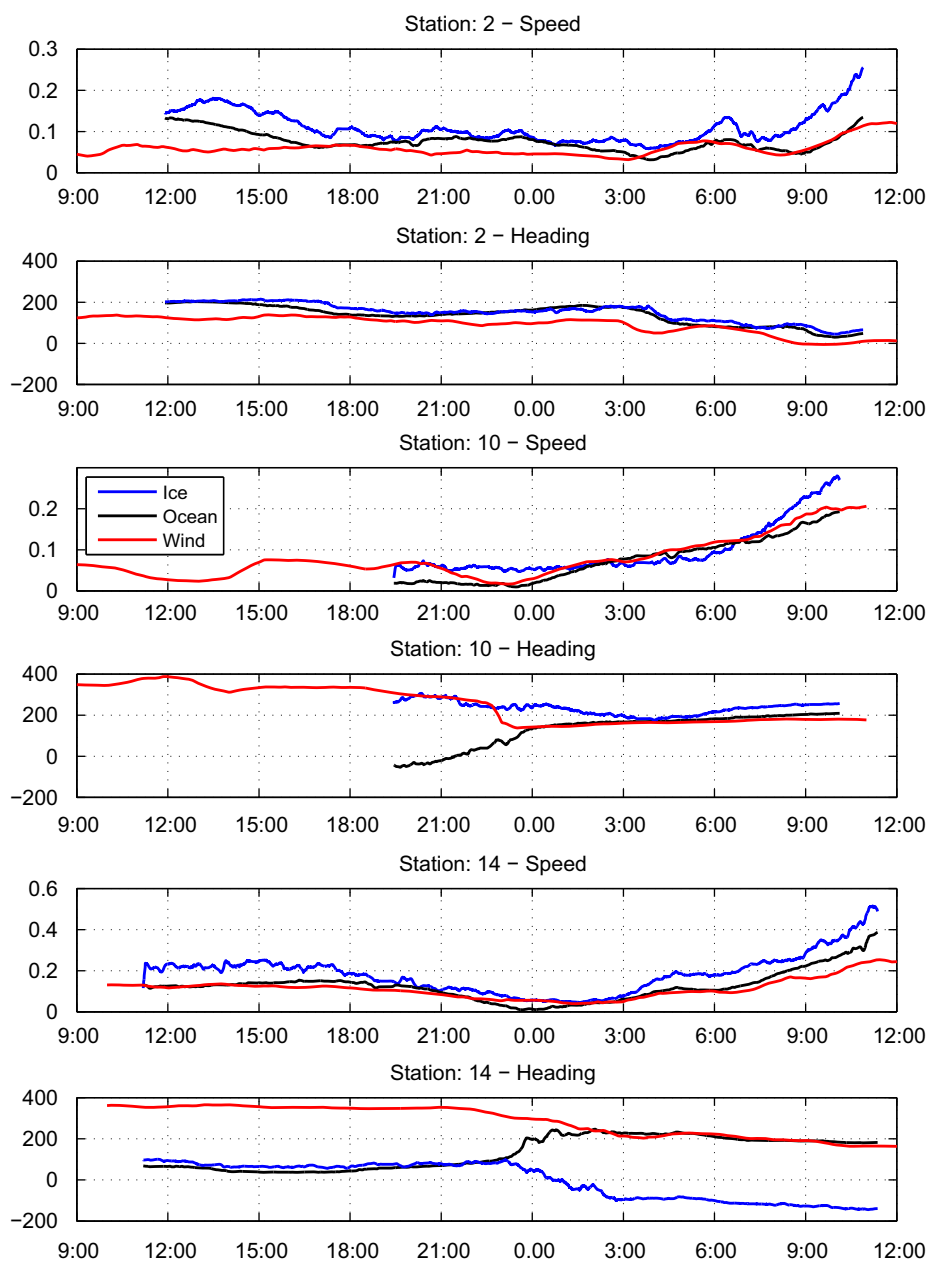

Fig. 13. Time series of ice drift (blue lines), ocean current (black lines) and 1% of the 10 m wind speed (m s^{-1}, red lines) and relative heading (degrees) for stations 2, 10 and 14 (see Fig. 10). Time shown as local time from 9.00 am to 12.00 pm the following day. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

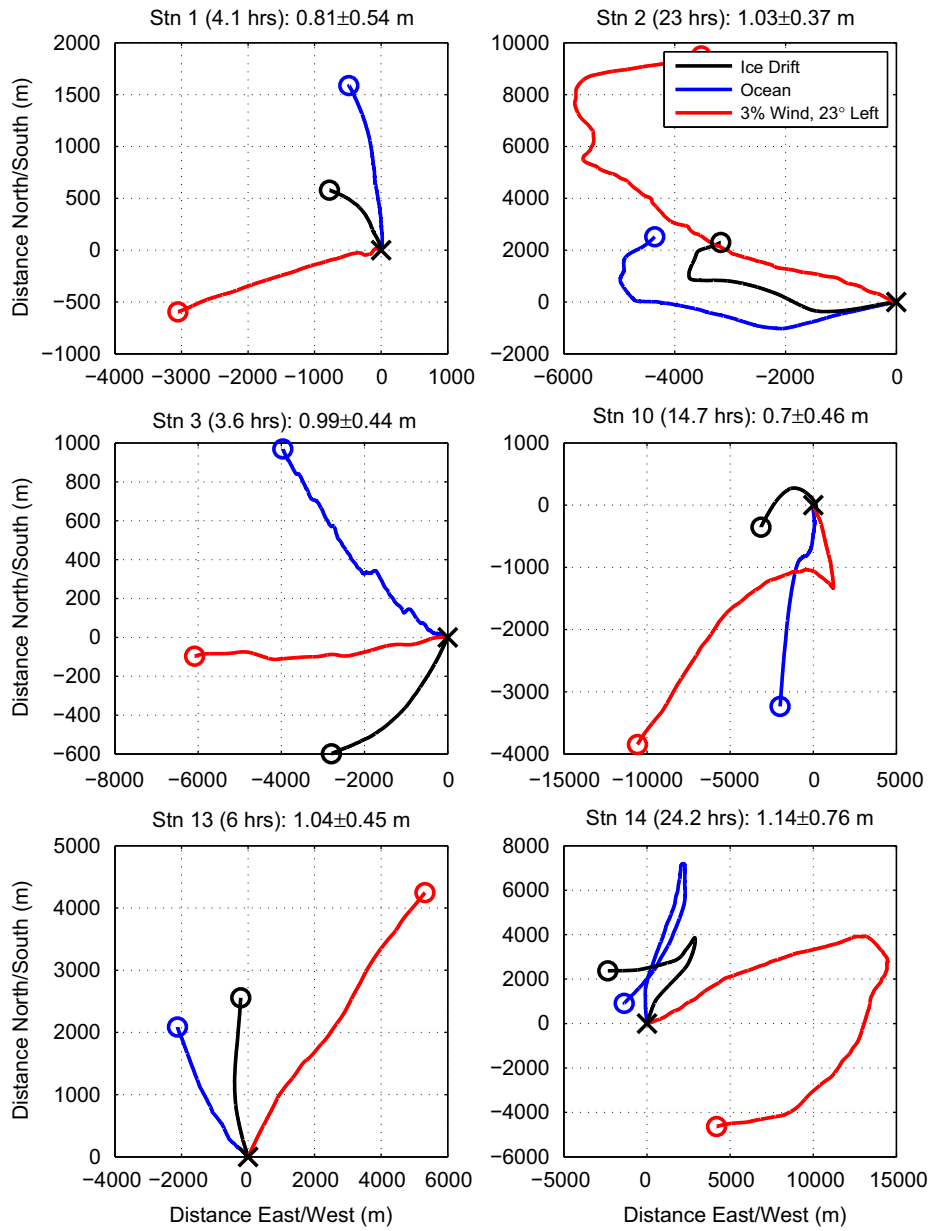

Fig. 14. Progressive vector diagrams of ice drift (black), wind (red), ocean current (blue) and combined ocean/wind (green) for stations 1–3, 10, 13–14 as labelled with period (h). The wind tracer is calculated using 3% of the observed wind speed rotated left by 23° following Martinson and Wamser (1990). The *x* and *y* axes are the cumulative distance (m) travelled in the east/west and north/south directions, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Apart from these few instances, however, the ocean and wind generally have very similar headings and trends.

Fig. 14 provides a qualitative view using progressive vector diagrams for ice, ocean and wind tracers at SIPEX stations 1, 2, 3, 10, 13 and 14. For the wind tracer we use the 'rule of thumb' of 3% of the wind speed, rotated 23° to the left following the results of Martinson and Wamser (1990). The ocean tracer represents 100% of the mean ocean current in the upper two ADCP bins (~ 3–19 m range). At no station did the 'rule of thumb' accurately predict the sea ice drift, in particular using the 3% value of the 15-m wind speed vector. At stations 2, 3 and 13 the result for the 'rule of thumb' would have improved by increasing the turning angle by 30–45°. Examining the mean ice thickness at the longer stations (2, 10 and 14), there is a qualitative increase in the influence of the ocean with increasing thickness. In Fig. 15 we present the principal component and correlation analysis for the ice drift, ocean current and wind time series shown in Fig. 13. At station 2, the percentage of variation explained by the 1st principal component of speed for ice/ocean and ice/wind pairs was 89% and 81%, respectively, with similarly large correlation pairs at the other stations. This demonstrates the significant degree of coupling between the wind–ocean system, and that as expected the wind substantially influences the surface ocean flow and sea ice drift on the time and spatial scales observed here. Future work will examine in more detail the short-term dynamics revealed by these observations with high-resolution wind data profile measurements directly above the ice.

5. Discussion

The most interesting and important water mass result from SIPEX was the observation of modified Circumpolar Deep Water at

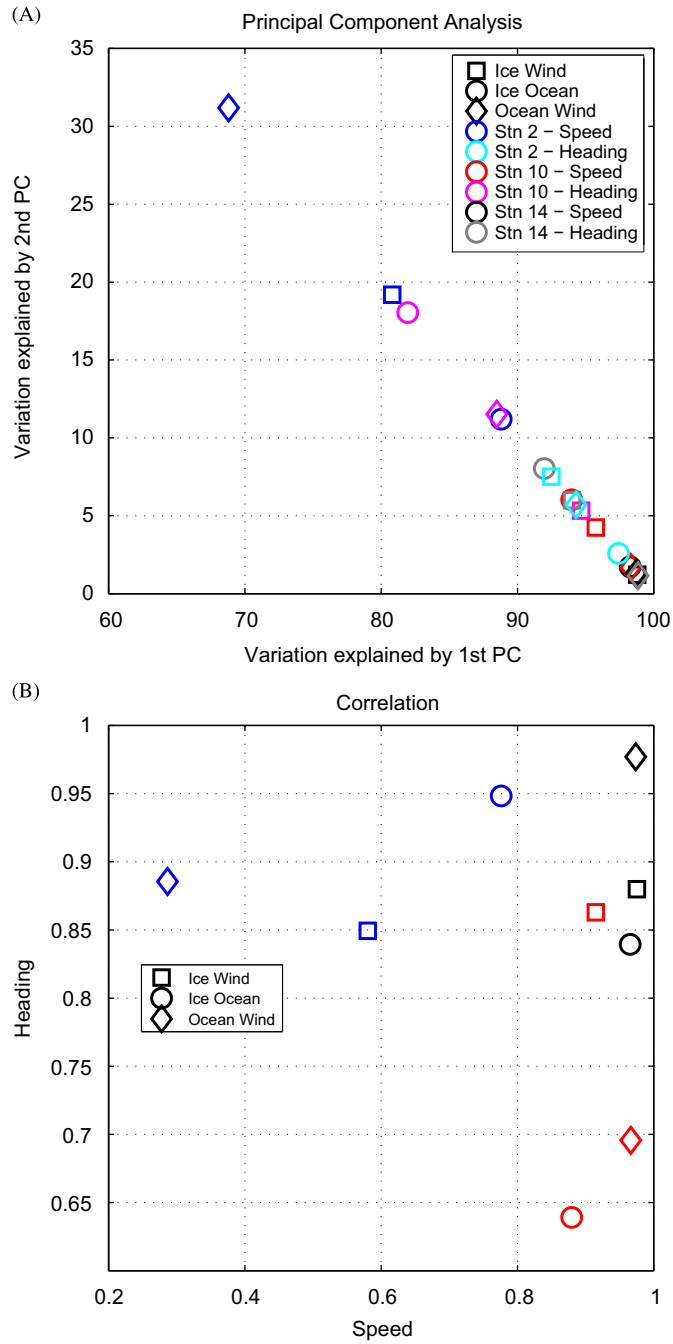

Fig. 15. Panel A. Principal component analysis. The *x* and *y* axes represent the variation explained by the 1st and 2nd principal components for (i) ice and wind (squares), (ii) ice and ocean (circles) and (iii) ocean and wind (diamonds) time series in Fig. 12. The analysis is for speed (blue, red and black symbols for stations 2, 10 and 14) and heading (cyan, magenta and gray for stations 2, 10 and 14). Panel B. Correlation analysis. The *x* and *y* axes are for the speed and heading correlation. Colors and symbols as in panel A.

117°E warmer than 0 °C. Bindoff et al. (2000) reported similar intrusions of mCDW near this location in summer (January 1996) and noted that this was the primary location sampled during the BROKE survey between 80°E and 150°E to have this feature. Williams and Bindoff (2003) showed 'highly' modified Circumpolar Deep Water in the Adélie Depression in winter that crossed the shelf break at mid-depths above the denser shelf water below. Williams et al. (2008a) showed that these intrusions increased dramatically in summer after the sea ice production season. However, our result here finds mCDW that is much warmer and bears more similarity to shelf break exchanges in the western Ross Sea and Amundsen sea where the Antarctic Slope Front extends onto the continental shelf. Clearly, two observations of mCDW in the vicinity of each other over a 10-year period does not prove that these warm, saline intrusions have been constant over this time period. Nonetheless, this additional observation of mCDW in late winter suggests that these mCDW intrusions are not limited to the austral summer as in other locations and we speculate that this strengthens the case for these mCDW intrusions being persistent intra- and inter annually.

This is very relevant to the melting of continental ice in this region as mCDW provides the necessary heat flux. In addition to the strength of the mCDW intrusion in this region, the density of these intrusions relative to the local shelf water is also critical. To truly influence the melting of ice shelves, the mCDW must be dense enough to reach the grounding line. For the Mertz Glacier in the Adélie Depression, the mCDW intrusions occur at mid-depths and occupy the upper layer of the water column, providing some heat flux to the base of the floating glacier (300–400 m), but have little influence on the groundling line (> 1200 m). Here, the mCDW is the densest water mass occupying the bottom layer, implying that should it continue its path south, it could reach nearby grounding lines. Though the East Antarctic ice sheet is the most stable part of Antarctica, the nearby Totten and Moscow University IceShelf were reported by Rignot and Thomas (2002) to be in a state of negative mass balance. Most recently Pritchard et al. (2009) report that the glacial thinning in this region could be three times more than previously estimated. Enhanced ocean heat flux from a persistent, bottom-trapped inflow of mDCW would certainly play a major role in such changes. However, the exact bathymetry and oceanic pathways for these mCDW intrusions to reach the grounding lines in this region are not yet well known and further observational studies and bathymetry mapping are required.

There are two mechanisms that could contribute to mCDW being the densest water mass on the shelf in this location, despite its warm temperature. The first is the influence of cold, fresh Ice Shelf Water (ISW), from ocean/ice shelf interactions beneath the nearby Totten Glacier and Moscow Univsersity Ice Shelf, which is likely to provide a freshening feedback to the local shelf water properties. Secondly, the sea ice production in the nearby polynya regions may simply be insufficient to produce truly dense Shelf Water, often referred to as High Salinity Shelf Water. To examine this second point further we present the sea ice production for the polynyas identified in this region in Fig. 16. The Cape Poinsett and Dibble Iceberg Tongue polynyas produce $\sim 60\ km^3$ of sea ice each year. These values are relatively low in comparison with the total sea ice production in the Mertz Glacier and Cape Darnley polynya/dense shelf water formation regions ($\sim 180\ km^3$), though the region east of the Mertz Glacier has similar production to the Dibble ($\sim 60\ km^3$) and this has been shown to be sufficient to produce low salinity shelf water cascades resulting in Antarctic Bottom Water (Williams et al., 2010a). The Dalton Iceberg Tongue polynya is the next largest at $40\ km^3$, with the Voyeykov and Blodgett averaging $20\ km^3$ per year, and although they are persistent with relatively small interannual variability, they are most likely too small to generate sufficient dense shelf water for Antarctic Bottom Water, let alone in conjunction with negative feedback from the fresh water flux from nearby ice shelves.

However, we note that it has been shown in a modelling study by Kämpf (2005) that dense shelf water overflows and mCDW intrusions can be intrinsically linked. That is, cascades of dense shelf water over the shelf break can induce a compensating upwelling of mCDW. Following this idea, there may yet be dense shelf water formation, upstream or downstream of the mCDW intrusion at 117°E. The BROKE survey detected evidence of dense shelf water overflows and Antarctic Bottom Water (AABW)

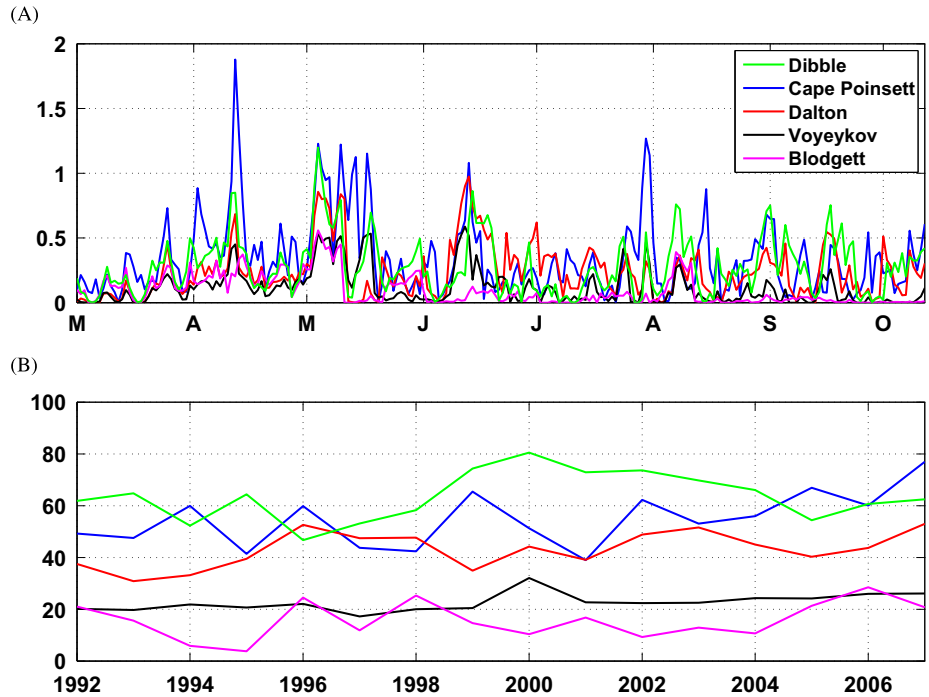

Fig. 16. Panel A. Daily mean sea ice production rates (cm day^{-1}). Panel B. Total sea ice volume (km^3) for the major polynya regions in Fig. 1. Data are calculated from ERA-Interim data.

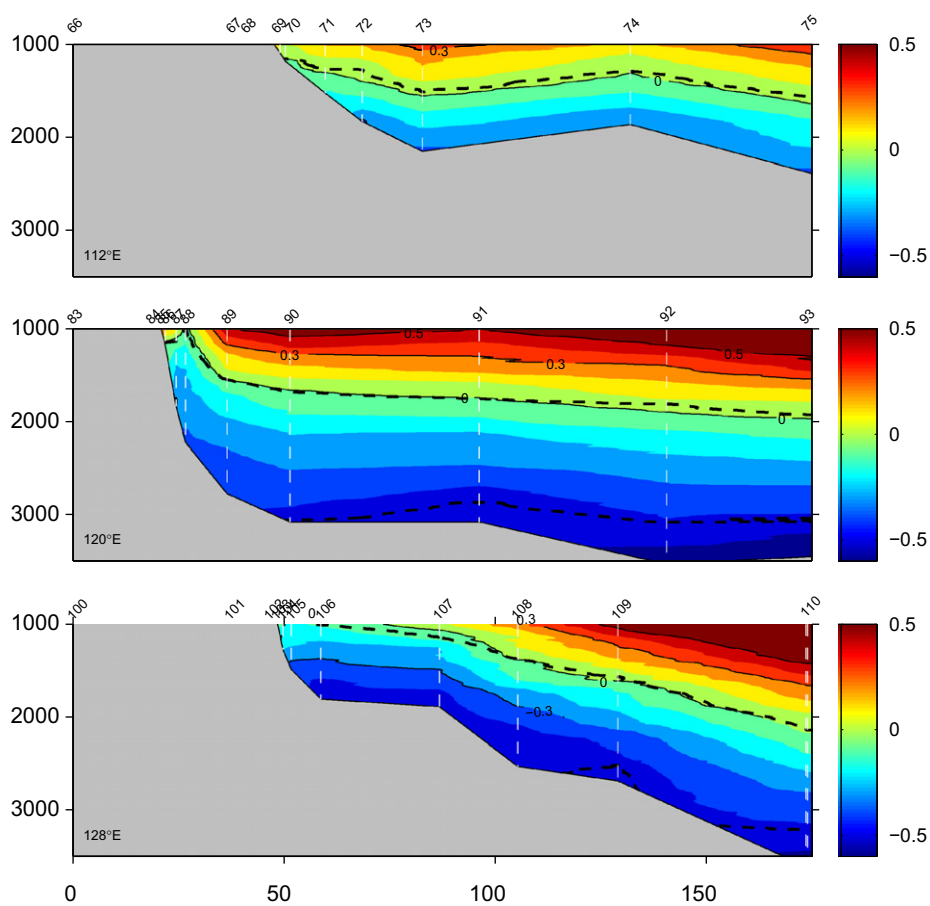

Fig. 17. As in Fig. 2, but for the deep and bottom layer between 1000 and 3500 m. The neutral density surface $\gamma^n = 28.27$ kg m^{-3} (black dashed line) indicates the upper limit of Antarctic Bottom Water (Orsi et al., 1995). From top to bottom the transects are located at 112°E, 120°E and 128°E, respectively.

production downstream from the Dibble Iceberg Tongue polynya in the elevated bottom layer concentrations of offshore CFC-11 (Williams et al., 2010a). We demonstrate this in Fig. 17, which shows the bottom layer in meridional vertical sections of potential temperature across the BANZARE and Sabrina coasts. The AABW layer, in particular the -0.3 °C isotherm shoals to 1500 m at 128°E to the west of the Dibble and then there is a subsequent decrease in temperature below 3000 m at 120°E. Bindoff et al. (2000) reported evidence of modified Shelf Water on the upper continental slope west of the Cape Poinsett polynya region. It is likely that given its modest amount of sea-ice production and the significant amount of melting occurring locally that there is very little dense shelf water being exported from the Dalton Iceberg Tongue Polynya.

As enticing as the concurrent measurements of sea ice drift, ocean currents and wind were for examining the comparative influence of wind and ocean on sea ice drift, the mixed results in Figs. 13–15 reflected the fact that our time series were simply too short to address this question conclusively. Similarities in the ocean and wind headings/velocities and overlapping correlation coefficients strongly indicate that the wind is important in driving the ocean, and hence directly or indirectly the sea ice too. However, as mentioned, this is only very short space/time scales, i.e. 24 h. Over longer timescales and in the presence of stronger ocean currents such as the Antarctic Slope Current, the ocean is clearly dominating, as indicated by the Ice-Argo drift pattern in Fig. 4 and reported by Heil et al. (2011) from drifting buoys. South of the shelf break, in the absence of strong currents, the synoptic wind patterns, tidal signals and clear inertial oscillations presented by Heil et al. (2011) show that sea ice ice drift is much more of a random walk than in locations where there are strong ocean currents. Ideally future deployments of this 'through-ice' ADCP system can be run for much longer periods and perhaps even modified to run autonomously over several months, as achieved in the study by Leonard et al. (2006) in McMurdo Sound.

5.1. Future work and challenges for observing winter oceanography in the SIZ

The pilot 'through-ice' oceanography program initiated on SIPEX was successful but can be improved in two key areas. Firstly the accuracy of the salinity measurements needs to be improved to facilitate greater comparison of results with more precise ship-based CTD observations. Secondly, the length of ADCP sampling must be increased to better understand the processes driving the intra-diurnal variability found during SIPEX. Perhaps the best result during SIPEX came from the Ice-Argo floats deployed on behalf of the University of Washington. The majority of these floats have continued to work over 2 years capturing two seasonal cycles. This dataset will greatly improve the understanding of the seasonal formation and partial decay of the winter mixed layer each year and will provide a comprehensive benchmark against which the current and future generations of climate models can compare with. However, the Ice-Argo floats remain limited to sampling the region north of the continental slope, that is north of the 3000-m isobath, and for this reason the 'through-ice' CTD system used during SIPEX is a relatively inexpensive and easy way to sample the regions on the upper slope and on the continental shelf break during sea-ice research voyages.

The biggest challenge for the next SIPEX-style research voyage is the effective wintertime sampling of the Antarctic Slope Front and Antarctic Slope Current region beneath the sea ice between the shelf break and the 1500-m isobath. We can report that the Antarctic Slope Current was 'experienced' by the R.V. *Aurora Australis* when heading south towards the continental shelf break when it began drifting within a 'river of ice' that was several tens of metres wide. The ship was in fact powerless to manuever efficiently due to the loose nature of the crushed sea ice within it and for several minutes found itself drifting helplessly west, less than a kilometre from numerous large icebergs grounded on the shelf break. Some modelling studies have found that the Antarctic Slope Current increases in winter beneath the sea ice (Mathiot et al., 2010), perhaps in response to the increased winds and the increased air/ice drag coefficient. More work is needed to understand the physical mechanisms behind this phenomena and its seasonal variability. However, *in situ* observations of the seasonal ice zone in winter, in particular the ASC, remain a challenge.

Acknowledgments

We would like to thank the officers and crew of the R.V. *Aurora Australis* and the Marine Science Support staff from the Australian Antarctic Division for their professionalism and support during the SIPEX voyage. The majority of this equipment used in the CTD system was previously used during the Amery Ice Shelf drilling project and Russell Brand constructed the overall system. The ADCP system was designed and built by the Marine Science Support group at the Australian Antarctic Division, with the loan of the ADCP kindly arranged by Mike Craven. Mark Rosenberg completed the post-processing of the FSI CTD data. Bathymetry data provided by Smith and Sandwell (1997). Ice-Argo floats deployed during SIPEX and presented in this paper were from the University of Washington and we would like to thank Steve Riser, Annie Wong, Dana Swift and Rick Rupin for their efforts in providing these data. These data were collected and made freely available by the International Argo Project and the national programs that contribute to it. (http://www.argo.ucsd.edu, http://argo.jcommops.org). Argo is a pilot program of the Global Ocean Observing System. This paper benefited greatly from the comments and suggestions of three anonymous reviewers.

References

Aoki, S., 2003. Seasonal and spatial variations of iceberg drift off dronning maud land, Antarctica, detected by satellite scatterometers. Journal of Oceanography 59, 629–635.

Argo, 2010. Argo home page. URL: 〈http://www.argo.ucsd.edu/〉.

Bindoff, N.L., Rosenberg, M.A., Warner, M.J., 2000. On the circulation and water-masses over the Antarctic continental slope and rise between 80 and 150° East. Deep-Sea Research II 47, 2299–2326.

Blackadar, A.K., 1982. The vertical distribution of wind and turbulent exchange in a neutral atmosphere. Journal of Geophysical Research 67, 3095–3102.

Cresswell, K.A., Tarling, G.A., Thorpe, S.E., Burrows, M.T., Wiedemann, J., Mangel, M., 2009. Diel vertical migration of Antarctic krill (*Euphasia superba*) is flexible during advection across the Scotia Sea. Journal of Plankton Research 31 (10), 1265–1281. doi:10.1093/plankt/ecss.fbp062.

ECMWF, 2009. ERA-Interim Reanalysis. URL: 〈http://www.ecmwf.int/research/era/do/get/era-interim〉.

ETOPO1, 2009. Global Predicted Bathymetry v11. Smith, W.H.F., Sandwell, D., 1997. Global seafloor topography from satellite altimetry and ship depth soundings. Science 277, 1956–1962. URL: 〈http://topex.ucsd.edu/marine_topo/〉.

Fahrbach, E., Rohardt, G., Krause, G., 1992. The antarctic coastal current in the southeastern weddell sea. Polar Biology 12, 171–182.

Heil, P., Allison, I., Massom, R.A., Worby, A.P., 2011. Effects of intensified atmospheric forcing and near-coastal currents on meso-scale sea-ice dynamics off East Antarctica. Deep-Sea Research II 58 (9–10), 1158–1171.

Heywood, K.R., Locarnini, R.A., Frew, R.D., Dennis, P.F., King, B.A., 1998. Transport and water masses of the antarctic slope front system in the eastern Weddell Sea. In: Jacobs, S., Weiss, R. (Eds.), Antarctic Research Series, vol. 75. American Geophysical Union, Washington, pp. 203–214.

Jacobs, S.S., 1991. On the nature and significance of the Antarctic Slope Front. Marine Chemistry 35 (1–4), 9–24.

Kämpf, J., 2005. Cascading-driven upwelling in submarine canyons at high latitudes. Journal of Geophysical Research 110, C02007. doi:10.1029/2004JC002554.

Kaneda, A., Takeoka, H., Koizumi, Y., 2002. Periodic occurence of diurnal signal of ADCP backscatter strength in Uchiumi Bay, Japan. Estaurine, Coastal and Shelf Science 55, 323–330. doi:10.1006/ecss.2001.0908.

Klatt, O., Boebel, O., Farhbach, E., 2007. A profiling float's sense of ice. Journal of Atmospheric and Oceanic Technology 24, 1301–1308. doi:10.1175/JTECH2026.1.

Leonard, G., Purdie, C., Langhorne, P., Haskell, T., Williams, M., Frew, R., 2006. Observations of platelet ice growth and oceanographic conditions during the winter of 2003 in McMurdo Sound Antarctica. Journal of Geophysical Research Oceans 111, C04012. doi:10.1029/2005JC002952.

Leppäranta, M., 2005. The Drift of Sea Ice. Springer-Praxis, Heidelberg, Germany.

Martinson, D.G., Wamser, C., 1990. Ice drift and momentum exchange in the winter Antarctic pack ice. Journal of Geophysical Research 95 (C2), 1741–1755.

Massom, R.A., Harris, P.T., Michael, K., Potter, M.J., 1998. The distribution & formative processes of latent heat polynyas in East Antarctica. Annals of Glaciology 27, 420–426.

Mathiot, P., Barnier, B., Gallée, H., Molines, J.M., Le Sommer, J., Juza, M., Penduff, T., 2010. Introducing katabatic winds in global ERA40 fields to simulate their impacts on the Southern Ocean and sea-ice. Ocean Modelling 35 (3), 146–160. doi:10.1016/j.ocemod.2010.07.001.

Meijers, A.J.S., Klocker, A., Bindoff, N.L., Williams, G.D., Marsland, S.J., 2010. The circulation and water masses of the Antarctic shelf and continental slope between 30 and 80°E. Deep Sea Research II 57 (9–10) 723–737. doi:10.1016/j.dsr2.2009.04.019.

Morales Maqueda, M.A., Wilmott, A.J., Biggs, N.R.T., 2004. Polynya dynamics: a review of observations and modelling. Review of Geophysics 42, RG1004/2004 2002RG000116

Muench, R.D., Padman, L., Gordon, A.L., Orsi, A., 2009. A dense shelf outflow from the Ross Sea, Antarctica: Mixing and the contribution of tides. Journal of Marine Systems 77 (4), 369–387. doi:10.1016/j.jmarsys.2008.11.003.

NODC, 2009. Operational Oceanography Group: Global Argo Data Repository. April 2007. U.S. Department of Commerce, National Oceanic and Atmospheric Administration, National Oceanographic Data Center, Silver Spring, Maryland, 2009. URL: 〈http://www.nodc.noaa.gov/argo/accessData.htm〉.

O'Brien, C., Virtue, P., Kawaguchi, S., Nichols, P.D., 2011. Aspects of krill growth and condition during late winter-early spring off East Antarctica (110 130e) during the SIPEX expedition (Sea Ice Physics and Ecosystem eXperiment September–October 2007). Deep-Sea Research II 58 (9–10), 1211–1221.

Orsi, A.H., Whitworth, T., Nowlin, W., 1995. On the meridional extent and fronts of the Antarctic Circumpolar Current. Deep-Sea Research I 42, 641–673.

Orsi, A.H., Wierderwohl, C.L., 2009. A recount of ross sea waters. Deep-Sea Research II 56, 778–795. doi:10.1016/j.dsr2.2008.10.033.

Padman, L., Fricker, H.A., Coleman, R., Howard, S., Erofeeva, S., 2002. A new tidal model for the Antarctic ice shelves and seas. Annals of Glaciology 34, 247–254.

Pritchard, H.D., Arthern, R.J., Vaughan, D.G., Edwards, L.A., 2009. Extensive dynamic thinning on the margins of the Greenland and Antarctic ice sheets. Nature, 1–5. doi:10.1038/nature08471.

Rignot, E., Jacobs, S.S., 2002. Rapid bottom melting widespread near Antarctic ice sheet grounding lines. Science 296, 2020–2023.

Rignot, E., Thomas, R.H., 2002. Mass balance of polar ice sheets. Science 297 (5586), 1502–1506.

Rosenberg, M., 2009. Sipex, Marine Science Cruise AU0701 - FSI CTD data processing. Technical Report, ACE CRC.

Sala, A., Azzali, M., Russo, A., 2002. Krill of the Ross Sea: distribution, abundance and demography of Euphausia superba and Euphausia crystallorophias during the Italian Antarctic Expedition (January–February 2000). Scientia Marina 66, 123–133.

Smethie, W., Jacobs, S., 2005. Circulation and melting under the Ross Ice Shelf: estimates from evolving CFC, salinity and temperature fields in the Ross Sea. Deep-Sea Research II 52, 959–978.

Tamura, T., Ohshima, K.I., Markus, T., Cavalieri, D., Nihashi, S., Hirasawa, N., 2007. Estimation of thin ice thickness and detection of fast ice from SSM/I data in the Antarctic Ocean. Journal of Atmospheric and Oceanic Technology 24, 1757–1772.

Tamura, T., Ohshima, K.I., Nihashi, S., 2008. Mapping of sea ice production for Antarctic coastal polynyas. Geophysical Research Letters 35. doi:10.1029/2007GL032903.

Thomson, A.F., Heywood, K.J., 2008. Frontal structure and transport in the northwestern Weddell Sea. Deep-Sea Research I 55, 1229–1251.

UW, 2009. University of Washington ARGO Profiling Drifters. URL: 〈http://flux.ocean.washington.edu/argo/info/general-info.shtml〉.

Whitworth, T., Orsi, A., Kim, S.-J., Nowlin, W.D., Locarnini, R., 1998. Water masses and mixing near the Antarctic Slope Front. In: Jacobs, S., Weiss, R. (Eds.), Antarctic Research Series, vol. 75. American Geophysical Union, Washington, pp. 1–27.

Whitworth III, T., Orsi, A.H., 2006. Antarctic Bottom Water production and export by tides in the Ross Sea. Geophysical Research Letters 33. doi:10.1029/2006GL026357.

Williams, G.D., Aoki, S., Jacobs, S.S., Rintoul, S.R., Tamura, T., Bindoff, N.L., 2010a. Antarctic Bottom Water from the Adélie and George V Land coast, East Antarctica (140–149°E). Journal of Geophysical Research 115, C04027, 29 pp. doi:10.1029/2009JC005812.

Williams, G.D., Bindoff, N.L., 2003. Wintertime oceanography of the Adélie Depression. Deep-Sea Research II, 1373–1392.

Williams, G.D., Bindoff, N.L., Marsland, S.J., Rintoul, S.R., 2008a. Formation and export of dense shelf water from the Adlie Depression, East Antarctica. Journal of Geophysical Research 113, C04039. doi:10.1029/2007JC004346.

Williams, G.D., Nicol, S., Aoki, S., Meijers, A. J.S., Bindoff, N.L., Iijima, Y., Marsland, S.J., Klocker, A., 2010b. Surface oceanography of BROKE-West, along the Antarctic margin of the south-west Indian Ocean (30–80°E). Deep-Sea Research II 57 (9–10), 738–757. doi:10.1016/j.dsr2.2009.04.020.

Williams, G.D., Nicol, S., Raymond, B., Meiners, K., 2008b. On the summertime mixed layer development in the marginal sea-ice zone off the Maw-son coast, East Antarctica. Deep Sea Research II 55 (3–4), 365–376. doi:10.1016/j.dsr2.2007.11.007.

Worby, A., Steer, A., Leiser, J., Galin, N., Yi, D., Allison, I., Heil, P., Massom, R., Zwally, H., 2011. Regional-scale sea ice and snow thickness distributions from in situ and satellite measurements over East Antarctica during the Sea Ice Physics and Ecosystem eXperiment (SIPEX). Deep-Sea Research II 58 (9–10), 1125–1136.